

AI Based Candidates Recommending System for a Company Vacancy

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Project Proposal Report

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DECLARATION OF CANDIDATE AND SUPERVISOR

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Date

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ABSTRACT

In the contemporary labor market, it is vital to assess a candidate's stress levels during interviews in order to have insight not just to candidates' ability to maintain composure, but how they are likely to perform in the changing, fluid contexts of workplaces. Our study, "Predicting Stress Levels from Eye Movement and Detection," is intended to develop and automate this assessment process using a range of current computer vision methods exploiting advanced AI. The study will organize visual information gained from the eye behavior of candidates during video interviews; stress is often shown through, blinking rates, gaze direction, pupil dilation, stability of eye movement, and so on.

This system utilizes eye detection models (e.g., Media pipe, OpenCV or deep learning-based gaze estimation pipelines) to detect and quantify eye features in real-time which are processed by machine learning models (e.g., Random Forest Classifier, Support Vector Machine (SVM), or any other classification method) to classify stress into high, moderate or low categories or similar, and provides a scoring mechanism that amalgamates them into a score quantifying a candidate's stress response in the interviews.

The initial findings suggest that applying computer vision methods and machine learning in an integrated real-time manner is effective. This research contributes to artificial intelligence-based recruitment through the provision of a scalable and objective assessment approach to broadly evaluate psychological resilience and emotional stability to provide a basis for data-driven hiring decisions for organizations.

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1. INTRODUCTION

Stress management and emotional stability are vastly important in today's highly competitive environments, particularly in job interviews, where the evaluator is looking at the candidate's knowledge but will also evaluate how the candidate handles the stress of being interviewed. Stress can impact communication, decision making, and overall performance, and should be evaluated by the recruiter. The classic interview, like most interview methods is heavily dependent on subjective human judgment which can be inconsistent and bias laden.

Recent development trajectories of artificial intelligence (AI), computer vision, and machine learning (ML) have allowed for the emergence of automated systems for real-time analysis of human behavior. In those automated systems, eye engagement analysis has emerged as a powerful, reliable proxy for psychological state. For example, the eye movements can include blinking frequency, pupil dilation, gaze direction, and movement of eye confidence have been identified and shown strong association to stress levels, giving measureable resources on each candidate's experiential emotional resilience.

This study considers real-time monitoring of eye engagement to infer stress, using techniques such as OpenCV and Mediapipe for eye landmark detection and machine learning classifiers like Random Forests and Support Vector Machines to classify stress levels. The eye engagement system uses eye characteristics throughout the in-person or remote interview to categorize stress levels into high, moderate, and low. With an objective, data driven system that will aid the recruiter in the evaluation process, the candidate can be assessed by additional feedback in tandem with the traditional evaluation process.

The study covers the design, development and evaluation of the system associated with the computer vision pipeline, feature extraction, and the predictive modeling. The goal of the system was an end-to-end system that measures eye engagement relevant patterns and categorizes them

into useful information about a candidate's stress, consequently being fairer, accurate, and more effective in ascertaining a superordinate purpose in recruitment.

1.1 Background Literature

Traditionally stress detection relied on physiological methods, such as heart rate variability, electro dermal activity, or electroencephalogram (EEG). In a study done by Healey and Picard (4), findings showed stress could be measured with multiple physiological sensors, but these methods relied on invasive sensors. While the models proved to be accurate in a controlled environment, they would not be useful for real-time situations such as an interview. Human stress is an evolved mechanism to respond to a threat, with physiological signals communicating the stress response. After the anticipation of a threat has removed the body can begin to perform a collection of reactions which could lead to changes in physiology (e.g heart rate). Devices are often too intrusive to reveal any meaningful information in non-contact or emergent situations.

Camera-based detection systems of stress are now available to help counter these limitations. It has been shown that eye-related features (for example, blink rate, gaze stability, saccadic movement, and pupil dilation) all correlate well to stress levels. Specifically, Ji et al. [2] showed that eye-tracking technology could be used to evaluate cognitive load and stress during task performance with effort to make non-invasive detection easier, resulting in significant levels of effortlessness. Camera-based systems would need a camera device (e.g. webcam), which is much easier to acquire and use in real-world contexts. Greater usability makes them more valid systems in terms of their real world applications.

However, the more recent or new area of research that continues to enhance stress-detection systems is the incorporation of machine learning (ML) with computer vision resources. For example, current computer vision frameworks such as OpenCV and Mediapipe allows researchers to extract eye and facial landmarks with great detail in real time. Abdrabou et al. [5], used convolutional neural networks (CNNs) to predict stress using eye and micro-expression features, demonstrating CNN based approaches perform far better than regulated pre-designated features and hand-coded machine-learning representations of images/pre-processed images, while describing the advantages of mappers or descriptors from various labelled datasets as more

automated features which would define uniform discriminative capabilities of subjects while avoiding biases or features based on the differences between individuals and/or environments.

Hybrid methods that combine eye features with additional modalities also exhibited accuracy. Sriram et al. [7] combined gaze tracking data with speech features in predicting stress in interview settings. Yin et al. [9] integrated facial expressions, eye engagement, and physiological sensors to investigate multimodal fusion, and found that multimodal fusion minimizes false positives, increases reliability under various conditions.

Stringency and real-time detection are contextual details of stress prediction systems. Kaur et al. [6] pointed out that efficient algorithms needed to process eye-tracking and classify the stress level without the possibility of being interrupted and delayed even a couple of seconds were not ideal. Stress detection stems from extreme situations, such as interviews and other high-stress workflows. The interviewer now can monitor stress levels and gain varying degrees of understanding of the candidate's emotional state and resilience in real-time.

Using machine learning classifiers has significantly improved the accuracy of stress predictions. Stress has been classified using Support Vector Machines (SVMs), Random Forests, and deep neural networks. Wu et al. [8] provided an SVM-based gaze analysis framework for detecting stress, and it achieved high accuracy in a controlled environment. More recent developments have explored various models using architectures that can explore or represent time between blink movements, and further research will build on these insights.

The novelty of eye engagement-based stress prediction is that it presents a way to monitor candidates throughout the interview, while being discrete, instantaneous, and scalable. By leveraging computer vision, eye landmark detection, and machine learning classifiers, it is possible to provide recruitment teams with actionable insights without sensors obstructing their

view. Consequently, this may lead to fairer, more data-driven assessments that are less biased and subjective than traditional recruitment practices.

As the field develops, future research will likely consider multimodal integration, self-learning AI models, and further robustness across cultural and environmental differences, thus allowing the eye engagement–based stress prediction system to become more robust, adaptive, and widely deployable.

1.2. Research Gap

While eye-tracking and gaze estimation technologies have made notable strides, there are still a number of gaps in the literature that prohibit the usefulness of eye engagement detection, namely in robot-based behavioral analysis at the interview stage. The identified gaps seem to stem from issues with gaze accuracy, head pose variability, lighting, real-time processing, and how gaze was subsequently interpreted as a marker of stress levels. Each subsequent subsection dives deeper into these major gaps relating to performance for eye engagement research.

For most of the literature identified, the tracking task is primarily of general-purpose use with regards to gaze tracking and eye movement detection. However, there continues to be little research that is focused on identifying whether the user is specifically looking into a camera during a live video-based interaction in a greater way. Current work pieces, like Zhang et al. [1], and Chen et al. [3], specify using a pre-trained gaze estimation model but there are simply too many confounding factors of the user, i.e. their camera angle, screen position, and body posture will create errors of great variability in unconstrained contexts like online interviewing. This evidence would suggest the need for more specific algorithms that are able to respond to user context changes and could reliably ascertain better direct camera engagement.

Another void in the literature surrounds the relationship between eye engagement and stress detection. While some studies (e.g. Wang et al. [6]) show correlations between gaze aversion and stress or anxiety, much of the literature is focused on physiological signals (e.g. heart rate, voice tone, facial micro-expressions, etc.) in regards to stress prediction. Very few studies have sought out eye engagement as a primary feature for estimating stress-level in interview contexts. This provides a more definitive opportunity to build off existing behavior-analysis systems to include indicators of stress using significant eye engagement detection.

There are also significant limitations in regards to environmental factors. Eye-tracking systems using RGB cameras often perform poorly in low-light settings, reflect off of candidate's glasses, or are not able to track the candidate's gaze consistently if they are moving their head frequently. While infrared-based gaze trackers provide even more robustness (and may work better in practice), they are unfortunately very expensive and impractical for large-scale virtual interview systems. More research is warranted with low-cost camera based methods, that maintain a high level of accuracy, for any potential use across a range of environments.

The gaze behavior area of human-site interaction remains unexplored. Whereas eye gaze indicates attention or stress, there is no research on how eye gaze engagement should be interpreted in structured environments (e.g., job interviews). For example, evidence of eye contact, looking away from the camera could mean stress, distracted, or just part of a natural pause in conversation. Systematic studies are required to engineer systems that not only reliably track gaze but can also contextualize engagement as it relates to candidate stress and performance.

To sum up, the current research gaps for eye engagement detection highlights issues that will require: (1) more precise gaze tracking techniques for direct camera engagement, (2) as a key feature of stress detection, gaze must be included, (3) robustness in different environment conditions, (4) real-time, light weight architecture, and (5) contextual interpreted eye engagement in interviews. All of these limitations will help facilitate the development of more accurate, scalable, lower cost behavioral analyses systems.

1.3. Research Problem

Selecting the right hire is one of the most daunting challenges that companies face today. In a traditional recruitment process, the three most common points of assessment are CV review, technical tests, and structured interviews. Defined recruiting relationships often miss critical elements of a candidate's psychological stability and emotional stress management. These are important facets of long-term job performance, strengths that be difficult to measure with traditional recruiting methods, in a saleable way.

Stress is one of the strongest drivers in the performance of candidate's during recruitment interviews. Some candidates may be high performers in technical skills yet demonstrate lower performance in the interview context. These applicants may be highly stressed while performing the interview. Whereas, candidates who are candidates that manage stress well are offered higher communication, problem solving, and adaptability skills than stressed candidates. Importantly, current recruitment systems lack a methodological and reliable way to quantify and understand a candidate's stress levels in real time and without too much intrusive measure.

Eye engagement is a set of behaviors consisting of levels of gaze, blink rate, fixation times and pupil dilation which has been shown to be valid indicators of stress and cognitive load. Candidate eye engagement provides a unique opportunity to analyses candidate stress levels using advances in computer vision/cognitive strain and machine learning applications. Eye engagement data could assess stress without the need to use intrusive wearable measures. However, despite its potential, multiple considerations are still need to be addressed before applying these measurements for recruitment-related purposes.

The key sub-problems are as follows:

(I) Lack of Non-Intrusive Stress Detection in Recruitment

Current recruitment platforms and recommendation systems rely heavily on resumes, psychometric tests, or manual observations by interviewers. These methods are either subjective or time-consuming.

(II) Limited Research on Eye Engagement in Candidate Evaluation

While eye-tracking has been widely used in areas like fatigue monitoring and attention studies, its application in job interviews for stress evaluation remains limited. Existing tools often do not connect eye engagement data to recruitment stress level of candidate.

(III) Accuracy and Real-Time Constraints

Stress detection through computer vision requires real-time performance to be effective during interviews. Challenges such as camera angles, lighting conditions, and obstructions (e.g., glasses) make it difficult to achieve accurate stress predictions in practical interview settings.

(IV) Integration with Candidate Recommendation Systems

Even when stress detection is achieved, it must be integrated into a recommendation framework. Recruiters should not only know whether a candidate was stressed but also receive insights into how stress levels might affect suitability for a role. Existing hiring systems do not provide this level of integration.

Therefore, the research problem lies in developing a **candidate recommendation system that integrates real-time, non-intrusive stress level prediction using eye engagement analysis**. The system aims to support recruiters by providing objective insights into a candidate's stress response, thereby improving the fairness, reliability, and accuracy of hiring decisions.

1.4 Objectives

1.4.1 Main Objective

The purpose of the Eye Engagement Analysis System is to enhance behavioral analysis in job interviews by tracking, identifying, and visually analyzing the eye movements and gaze patterns of interview candidates in real-time. Eye engagement is a significant nonverbal indicator of attention, confidence, honesty, and the ability to actively engage in the interview. However, subjective human assessment is at the center of traditional interview methods, which can result in emotional bias, inconsistencies, and potential in competencies in the evaluation of job candidates.

This system aims to objectively analyze the eye behaviors of candidates in terms of where a candidate's gaze rests, how long they pace their gaze on a particular outcome, how often they blink, and how often they refocus or switch from one visual anchor in the interview to another using advanced computer vision (CV) and machine learning (ML) techniques. The ability to continuously assess the candidate's eyes will provide researchers with an in-depth understanding of whether a candidate remains positively engaged, remains engaged in the interviewer, or whether attention in the context of eye engagement is shifting towards one or multiple environmental stimuli outside of the interview.

1.4.2 Specific Objectives

To achieve the main objective of enhancing behavioral analysis in interviews through eye engagement monitoring, the system will fulfill the following specific objectives:

• **Real-Time Eye Movement tracking**

The first key objective is to develop a real-time eye tracking mechanism that continuously monitors and records the candidate's eye movements throughout the interview session. High-resolution cameras, placed at optimal angles, will capture live video streams of the candidate's facial features. These streams will then be processed using computer vision algorithms such as facial landmark detection and pupil localization to accurately identify the direction and position of the eyes.

The system will analyze eye fixation points to determine where the candidate is focusing—whether it is the interviewer, the environment, or irrelevant distractions. Real-time monitoring ensures that engagement levels are captured instantly without delays, allowing the interviewer or automated analysis module to gain immediate insights into the candidate's attentiveness and behavioral patterns.

This objective is critical as it forms the foundation of the entire analysis system, enabling reliable measurement of engagement metrics during dynamic and natural interview conditions.

• **Quantitative Engagement & Confidence Measurement**

The second specific objective is to transform raw eye movement data into measurable indicators of engagement and confidence. Traditional interviews often rely on subjective perceptions such as “*the candidate seemed confident*” or “*the candidate avoided eye contact.*” To eliminate this subjectivity, the system will introduce quantifiable metrics such as:

- Average fixation duration on the interviewer's face.
- Percentage of interview time spent maintaining confident eye contact.
- Frequency of gaze shifts towards distracting elements.
- Blink rate variations under pressure.

By combining these metrics, the system will compute an **Engagement & Confidence Index**, a numerical score representing the candidate's overall attentiveness and confidence level during the interview. This quantitative approach ensures that evaluations are objective, consistent, and comparable across multiple candidates.

- **Enhance Accuracy in Different Interview Conditions**

Another critical objective is to ensure that the system performs reliably under diverse interview settings and environmental conditions. Factors such as poor lighting, reflections from glasses, background distractions, and differences in eye shapes or skin tones can affect the accuracy of eye tracking systems.

To overcome these challenges, advanced algorithms will be implemented to normalize variations in lighting and facial features. Adaptive thresholding techniques will be applied to ensure consistent eye detection in low-light or high-glare scenarios. Furthermore, the system will be trained on diverse datasets to enhance robustness across candidates of different ethnicities, genders, and facial characteristics.

By ensuring high accuracy in various real-world interview environments, the system will remain reliable and unbiased, delivering consistent results under all conditions.

- **Integration with Multimodal Behavioral Analysis**

The next objective is to integrate eye engagement and confidence data with other behavioral indicators, such as speech tone, facial expressions, and body posture, creating a multi-dimensional analysis framework. While eye engagement is a strong indicator of attentiveness and confidence, it becomes more powerful when combined with additional non-verbal cues.

For example, if a candidate maintains eye contact but simultaneously shows hesitations in speech or nervous body language, the system can highlight discrepancies that suggest lack of confidence. Conversely, consistent eye engagement coupled with steady voice tone and positive facial expressions would indicate a high level of confidence and preparedness.

This multimodal integration will allow recruiters to obtain a holistic view of candidate behavior, reducing the risks of over-reliance on a single parameter and improving the accuracy of final assessments.

• **Intelligent Feedback and Reporting System**

The final objective is to develop an intelligent reporting mechanism that presents eye engagement and confidence insights in a clear, understandable, and actionable format. After each interview session, the system will generate a comprehensive report containing:

- Visual heat maps showing gaze concentration areas.
- Time-series graphs of engagement and confidence fluctuations during the interview.
- A summary of key behavioral patterns detected.
- An overall Engagement & Confidence Index with supporting analysis.

In addition, the system will provide real-time alerts for interviewers, such as indicating when a candidate avoids eye contact for prolonged periods, appears distracted, or demonstrates consistent confidence through strong visual engagement.

This reporting system ensures that the data collected is not only accurate but also meaningful and easy for recruiters to interpret, empowering them to make fair, data-driven decisions during candidate evaluations.

2. Methodology

In this section, we present an overview of the methodology used in the design and development of the candidate recommendation system that predicts stress based on eye engagement. The intention is to assist companies and other organizations make hiring decisions and objectively evaluate the stress responses of the candidate in a job interview context. The candidate recommendation system provides the company or organization some actionable insights, to assist HR and talent decision-making by identifying candidates for particular roles that will be a better fit, provide organizations with a better quality of recruitment by critically evaluating their talent pipeline and reduce any biases in hiring processes.

The system takes input from a number of advanced technologies to meet its objectives. Machine learning algorithms are used to determine patterns in candidate responses and then determine correlations to elements of stress. Computer vision processes are utilized to watch eye engagement and collect data using real-time data feeds while analyzing eye engagement data in real time. Data analytics is used to accept the data and convert raw eye engagement data into interpretive metrics that can reflect the candidates stress level.

The methodology is outlined in a number of steps including data collection, data preprocessing, feature extraction, model training, and real-time analysis during the interview process. The use of these technologies and development methods/integrated approach strongly provides a strong and sufficiently valid measure of candidates' stress levels, improving the process for recruiters and HR professionals.

2.1 Eye Engagement and Stress Level Detection Setup

The proposed system is composed of two major processing units:

- **Eye Engagement Detection Unit** – Using camera modules, real-time video feeds of the candidate are captured during interviews. These feeds are processed using computer vision techniques (OpenCV, MediaPipe) and deep learning models to track eye movement, gaze direction, and blink rate. The data is used to measure attention span, focus, and possible stress indicators.
- **Stress Prediction Unit** – Extracted eye engagement features are processed through a trained machine learning model (e.g., Random Forest, CNN, or LSTM). The model maps eye behavior patterns (like abnormal gaze shifts, frequent blinking, or reduced eye contact) to corresponding stress levels (low, moderate, high).

The processed data is uploaded to a cloud-based server, where the results are analyzed and stored. A web-based dashboard displays candidate performance, including engagement metrics and predicted stress levels, which are then incorporated into the final recommendation system for employers.

2.1.1 Requirement Gathering

The requirement gathering for the Candidate Recommendation System using eye engagement was conducted through literature reviews, interviews with HR managers, and analysis of existing recruitment tools. The objective was to identify limitations of current systems and create an intelligent, affordable, and accurate solution for candidate evaluation.

1. Review of Existing Recruitment Systems

Current recruitment tools primarily focus on CV filtering, aptitude tests, and verbal interviews. However, they often lack psychological and behavioral analysis, especially non-verbal cues such as eye engagement. This creates a research gap where behavioral stress analysis can enhance fairness and accuracy in hiring decisions.

2. Challenges in Stress Detection from Eye Engagement

Detecting stress from eye engagement alone presents challenges such as variation in lighting, differences in individual behavior, and camera quality. To overcome these, preprocessing techniques (face alignment, eye region extraction) and noise reduction filters are used. Furthermore, combining multiple features (blink rate, gaze fixations, saccades) improves accuracy.

3. Integration with Recommendation System

The system does not stop at predicting stress but integrates results into a recommendation engine. Candidate profiles are enriched with stress analysis reports and engagement metrics. Recruiters can view an overall ranking of candidates along with traditional evaluation metrics, allowing them to make data-driven hiring decisions.

4. Simulations and Test Feedback

Prototype testing was conducted with mock interview sessions where participants were recorded under controlled conditions. The system successfully detected variations in stress levels. Feedback from HR professionals suggested the importance of presenting results in a simple, visual format (graphs, stress index scores) and avoiding false alarms. Based on this feedback, stress detection thresholds and data visualization techniques were refined.

2.1.2. Gathering and Analyzing Requirements

The system requirements were divided into functional and non-functional to execute the design's installation and evaluation stages for the Eye Engagement–Based Stress Level Prediction component of the Candidate Recommendation System.

2.1.2.1 Functional Requirements

- Detect and track eye engagement in real time:
The system must continuously capture and analyze candidate eye movements during interviews using computer vision techniques to measure attention levels.
- Classify engagement and stress levels:
Based on eye activity (e.g., blink rate, gaze direction, fixation duration), the system must classify candidates into engagement categories and predict potential stress levels.
- Integrate engagement metrics with candidate evaluation system:
The predicted stress/engagement scores must be synchronized with the recommendation engine to provide recruiters with a comprehensive assessment.
- Store candidate engagement data securely in the database:
All processed data must be saved in a secure backend to allow future retrieval, auditing, and analysis by hiring managers.
- Provide recruiter with real-time dashboard feedback:
The system should display stress-level indicators, engagement graphs, and summary reports in real time during the interview session.

2.1.2.1 Functional Requirements

- Latency must be below 2 seconds for stress predictions:
The time from video frame capture to engagement/stress-level prediction should not exceed 2 seconds to ensure real-time evaluation.
- System should work under different lighting and candidate conditions:
The eye detection module must remain reliable in diverse environments (low light, glare, and camera quality differences) and across candidates with different facial/eye features.
- Optimized for continuous monitoring:
The system must handle long interview sessions without overheating, lagging, or excessive resource consumption.
- Data privacy and security must be guaranteed:
Candidate video feeds and stress predictions must be encrypted and handled according to ethical and legal standards (e.g., GDPR-compliance where applicable).
- Recruiter interface must be clear and intuitive:
The dashboard should use simple indicators (e.g., color coding, engagement score percentages) so non-technical recruiters can quickly interpret candidate stress levels.
- Scalability of the system:
The framework must support multiple simultaneous interview analyses without degrading prediction accuracy or response time.

2.1.2 System Architecture

The focus of this project is on the design and development of a Real-Time Eye Engagement and Stress-Level Prediction Component that will be integrated with a Candidate Recommendation System. The goal of the project is for employers not only to evaluate the candidate's technical competencies but also to analyze behavioral and psychological attributes during an interview process.

The architecture of the system consists of five main hardware and software modules: Eye Detection and Tracking, Engagement and Stress Predictions, Data Fusion with Recruitment Metrics, (Secure) Cloud Communication and Visualization on the Recruiter Dashboard. The system that will be developed will capture the workflow so that the candidates behavioral states are analyzed appropriately and the recruiters have meaningful insights to help them in their hiring decisions.

➤ Eye Detection and Tracking

• Data Collection:

The system initiates with real-time video acquisition from a candidate during an interview session. Input is captured using a standard webcam, laptop camera, or mobile device camera, ensuring accessibility without requiring expensive hardware. The video stream is sampled into frames, which are preprocessed (resized, normalized, grayscale conversion) before being fed into the detection model.

• Model Architecture:

The detection component leverages a Convolutional Neural Network (CNN) trained on labeled datasets of eye states (open/closed) and gaze positions. This enables accurate localization of the eye region from the candidate's face and detection of subtle changes in eye dynamics. The CNN model achieved a final training accuracy of ~99% and a validation accuracy of ~93%, ensuring reliable real-time performance.

- **Real-Time Tracking:**

The model performs continuous blink detection, gaze estimation, and pupil movement tracking. Important behavioral metrics such as blink rate, eye closure duration, and fixation stability are extracted. These metrics provide a foundation for identifying whether a candidate is attentive, distracted, or experiencing stress during the evaluation process.

- Engagement and Confidence-Level Prediction

- **Feature Extraction:**

From the detected eye region, features such as blink frequency, saccadic movement patterns, gaze deviation, and pupil dilation are continuously extracted. These features are standardized and streamed into the classification pipeline.

- **Predictive Analysis:**

A deep learning-based classification model processes these features to categorize candidate states into:

- **Engaged:** Normal blink rate, stable gaze, high fixation duration.
- **Distracted:** Frequent gaze shifts, reduced fixation, irregular blink intervals.
- **Confident:** Steady gaze, balanced blink rate, controlled pupil movement, longer fixation on interviewer.

Engagement and confidence are quantified into numerical scores that reflect the candidate's behavioral state throughout the interview. The classification helps recruiters identify candidates who remain composed and confident under pressure versus those showing signs of hesitation or disengagement.

- **Decision Logic:**

The system applies temporal smoothing to avoid false alarms, ensuring predictions are consistent over time rather than reacting to momentary fluctuations. This enhances reliability and minimizes noise in engagement analysis.

➤ Data Fusion and Integration

• Data Fusion:

The behavioral metrics generated by the engagement model are integrated with technical evaluation scores (such as coding test results, aptitude tests, or HR assessments). This fusion creates a multi-dimensional candidate profile that reflects both cognitive and psychological performance.

• Holistic Evaluation:

The integrated system applies weighted scoring to combine:

- Technical skills and knowledge
- Engagement level consistency
- Stress-level analysis
- Soft-skill indicators from behavioral metrics

This ensures that final recommendations are balanced, capturing not only what the candidate knows but also how they perform under real-world stress conditions.

➤ Secure Cloud Communication

• Data Transmission:

Engagement and confidence data are continuously transmitted from the local processing unit to a secure cloud server for storage, large-scale analytics, and recruiter access. To ensure data integrity and confidentiality:

- **Authentication:** Cloud endpoints require **multi-factor authentication (MFA)** and **API keys with role-based access control (RBAC)** to prevent unauthorized access.
- **Data Integrity Checks:** Hashing algorithms (SHA-256) verify that no data has been tampered with during transmission.

- **Reliability:**

The system is optimized to deliver near real-time insights during interviews with minimal delays.

- **Low Latency:** Ensuring recruiters receive immediate feedback without disrupting the flow of conversation.
- **Redundancy:** Automatic failover mechanisms reroute traffic through backup servers if the primary communication channel fails.

- **Scalability and Compliance:**

- The cloud architecture supports horizontal scaling, enabling multiple candidate interviews to be processed simultaneously without performance degradation.
- Data storage and handling comply with GDPR and other data protection regulations, ensuring candidates' personal and behavioral information remains private and ethically managed.

➤ **Recruiter Dashboard and Notifications**

- **Real-Time Dashboard:**

The recruiter interface is designed to be **intuitive, visual, and actionable**, giving recruiters immediate insights during live interviews. The dashboard presents:

- **Engagement Levels:** Displayed as a **percentage graph over time**, showing how attentive the candidate remains during different interview segments.
- **Confidence Scores:** Categorized into **low / moderate / high**, represented through numeric values and visual indicators (e.g., confidence meter or gauge).
- **Engagement vs. Performance Correlation:** Graphical comparison of candidate engagement/confidence with their performance (e.g., technical answers, problem-solving scores). This helps recruiters understand whether dips in engagement/confidence correlate with weak responses.

- **Trend Analysis:**

Recruiters can analyze **historical trends** throughout the interview session:

- **Engagement Graphs:** Visual plots show fluctuations in attention levels, such as steady engagement during introductions versus noticeable drops during detailed technical discussions.
- **Confidence Tracking:** Timeline graphs highlight moments when confidence rises (e.g., when answering familiar topics) or declines (e.g., under challenging scenarios).
- **Pattern Recognition:** Helps recruiters identify **trigger points**—for instance, if a candidate consistently loses focus during long explanations or demonstrates reduced confidence under pressure questions.

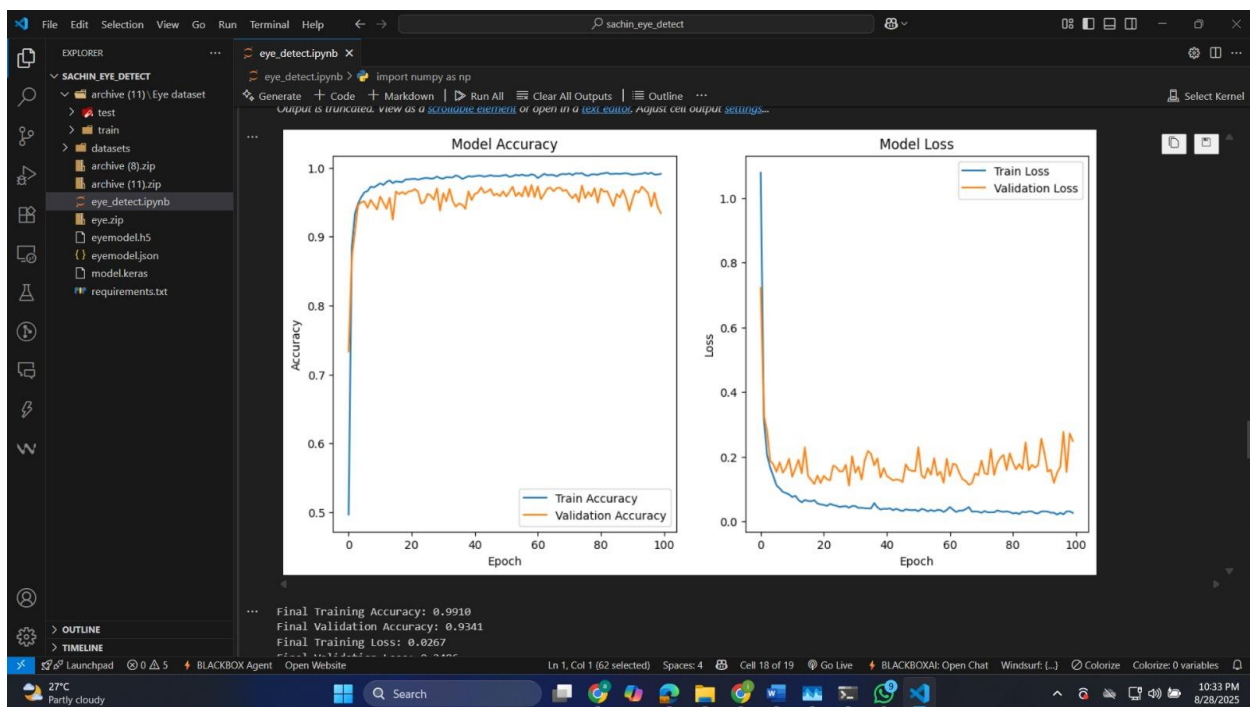
- **Final Candidate Recommendation:**

At the end of the interview session, the system generates a **comprehensive report** combining:

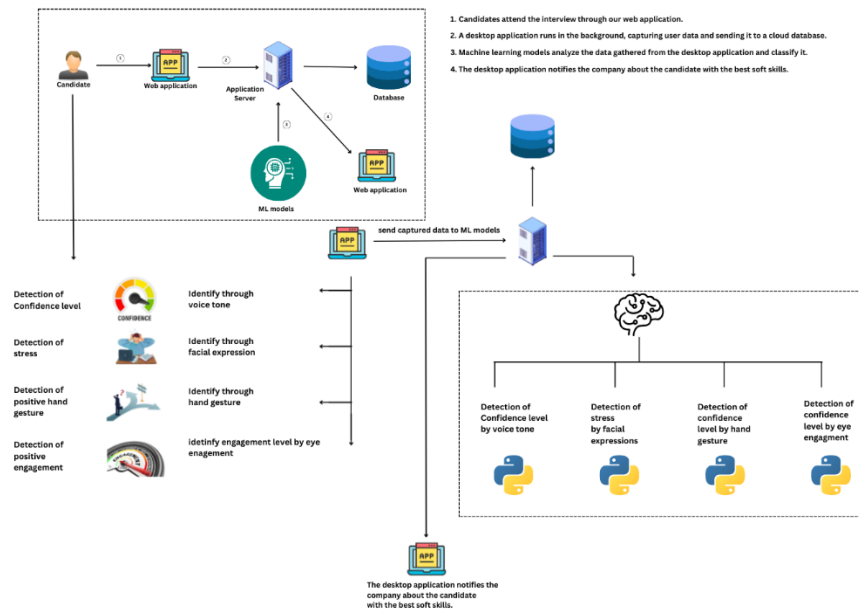
- **Technical Assessment Scores** (knowledge and problem-solving ability)
- **Engagement Metrics** (average engagement percentage, attention span, focus duration)
- **Confidence Metrics** (average confidence score, fluctuations, recovery under pressure)

This final recommendation assists recruiters in making data-driven hiring decisions, ensuring candidates are evaluated not only on their knowledge but also on their composure, focus, and confidence during interviews.

The System Architecture for the Eye Engagement and Stress-Level Prediction Element consists of a CNN-based eye tracker, extraction of behavioral features and classification of stress, secured communication, and visualization for the recruiter. The integration of technical evaluation with psychological insight provides recruiters with an entirely comprehensive, real-time, and objective way to evaluate candidates in the job market. This allows companies to not only find qualified workers but also identify candidates who can perform under pressure, stay engaged while working on a task, and be resilient when the workplace changes.



2.1.4 System Diagram



The Candidate Recommendation System builds upon a combination of multiple AI-driven modules to evaluate candidates in ways beyond their technical skills. The system leverages behavioral signals such as eye engagement, level of stress, tone of voice, hand gestures, and facial expressions during interviews, which are then processed via ML models to provide a full candidate recommendation.

➤ Direct Gaze Detection

- The system continuously analyzes the candidate's eye orientation to determine if they are looking directly at the camera.
- **High Confidence:** The candidate consistently maintains eye contact, showing engagement, focus, and composure.
- **Low Confidence / Distracted:** The candidate frequently looks away from the camera, indicating nervousness, distraction, or lack of self-assurance.

➤ **Gaze Duration**

- Measures how long the candidate maintains eye contact in each instance.
- **High Confidence:** Longer, stable periods of looking at the camera reflect attentiveness and comfort in the interaction.
- **Low Confidence:** Short, intermittent glances or persistent avoidance of the camera suggest hesitation or insecurity.

➤ **Frequency of Gaze Aversion**

- Tracks the number of times a candidate looks away from the camera during the session.
- **High Confidence:** Minimal gaze aversion, maintaining consistent visual engagement.
- **Low Confidence:** Frequent shifts of gaze away from the camera, possibly indicating anxiety or lack of focus.

Machine Learning-Based Analysis

An **Eye detection model** processes video frames to classify whether the candidate is looking at the camera or away at any given moment. These observations are aggregated over time to produce a confidence profile, which identifies trends in eye contact behavior:

- **High Confidence:** Majority of the time, eyes are directed at the camera, showing composure and engagement.
- **Moderate Confidence:** Occasional looking away, but frequently returns gaze to the camera.
- **Low Confidence:** Persistent avoidance of camera eye contact, indicating discomfort or low self-assurance.

Confidence Scoring

The system generates a **numerical confidence score** based solely on direct eye contact metrics. This score reflects how often and for how long the candidate maintains visual engagement with the camera, providing a simple yet powerful measure of confidence.

Advantages of Camera-Focused Eye Engagement Analysis

- **Simple and Direct:** Focuses on one of the clearest indicators of confidence—eye contact.
- **Objective Measurement:** Eliminates subjective bias by quantifying gaze behavior.
- **Real-Time Feedback:** Detects moments of disengagement instantly, enabling dynamic understanding of candidate behavior.
- **Actionable Insights:** Helps recruiters identify candidates who are self-assured, attentive, and comfortable under observation.

By analyzing **direct camera engagement**, the system gives recruiters a straightforward, reliable measure of confidence. Candidates who maintain eye contact are likely to be confident, focused, and ready to communicate effectively, while those who avoid the camera may need further assessment for composure or engagement.

2.2 Commercialization Aspects of the Product

➤ Product Development and Refinement

Our candidate hiring platform is focused on measuring soft skill aspects, utilizing confidence detection through eye engagement as one of the applicable components. This component is meant to offer companies data-driven insight into a candidate's confidence level during interviews by analyzing real-time eye movement, gaze directions, and engagement levels.

This component is a computer vision based system that operates with a machine learning pipeline, using high-resolution cameras. The cameras capture facial data and eye data that we use advanced algorithms, with a combination of OpenCV and deep learning models, to process these images. These models are then able to extract key indicators of confidence, such as, eye contact duration, blink rates, and pupil dilation. The processed data is securely sent back to a backend in the cloud, where it can be visualized in a recruiter dashboard. The dashboard serves as a platform for immediate feedback and confidence scoring of each candidate that may be a valuable resource for the future.

Developing this module is a continuous process of developing and refining the algorithm into a machine learning model through pilot testing and real interviews, providing users with a tool that will assist them in their work. We leverage HR experts, behavioral psychologists, and recruiting managers, to obtain accurate and timely feedback when determining the robustness of our confidence predictions, correctly establish threshold analysis, and improving user interface design, as a group. Future iterations will also develop signaling behaviors to provide more robust confidence estimates, compromised by different lighting conditions, camera angles, and demographic differences of the candidate.

Much of the work done for development will include many GDPR related items, as well as local employment laws for data privacy regulations, to ensure that the candidate's data, is processed ethically and securely.

➤ Market Entry Strategy

Our confidence detection component addresses a growing demand for **data-driven hiring decisions**, appealing to corporate HR teams, recruitment agencies, and talent acquisition platforms.

Market entry will follow a dual approach:

- **B2B:** Targeting mid-to-large companies, recruitment agencies, and online interview platforms.
- **B2C:** Offering cloud-based tools to smaller enterprises and freelance recruiters.

Key launch initiatives include:

- **Online Platform Launch:** Accessible via our official website, with SaaS subscription options.
- **Partnerships:** Collaboration with recruitment software providers and HR consultancies to integrate the confidence detection module into existing hiring workflows.
- **Demo Sessions:** Live demonstrations at corporate HR events, career fairs, and training centers to showcase real-time candidate analysis.
- **Digital Marketing:** Targeted campaigns on LinkedIn, HR-focused forums, and social media, emphasizing data-backed hiring advantages.

➤ Pricing Strategy

Affordability and scalability are central to our pricing approach. The module will follow a **tiered SaaS subscription model**:

- **Basic Plan:** Core confidence scoring and visual dashboards for small teams.
- **Premium Plan:** Advanced analytics, historical trend tracking, and integration with ATS (Applicant Tracking Systems).
- **Enterprise Plan:** Custom packages for large corporations, including API access, multi-user dashboards, and personalized support.

Additional add-ons, such as **detailed behavioral reports** and integration with psychometric testing tools, will provide further value for enterprise clients.

➤ Distribution Channels

To reach a broad user base, the confidence detection component will leverage multiple distribution channels:

- **Direct Online Access:** Subscription-based access via our SaaS platform.
- **HR Software Integrations:** Collaboration with ATS and video interview platforms to embed the module.
- **Recruitment Consultants & Agencies:** Partnering with agencies to offer value-added services to clients.
- **Training Institutes & Universities:** Offering modules to HR and psychology programs for research and practical training.

Live webinars, workshops, and trial campaigns will be conducted to encourage adoption and generate awareness.

2.3 Implementation and Testing

2.3.1 Implementation

The Confidence Detection utilizing Eye Engagement module is developed to consider a candidate's non-verbal behavior within an interview and deliver real-time confidence scores to recruiters. The implementation of this feature utilizes a combination of computer vision techniques, machine learning models, and the cloud for data processing, resulting in a measured candidate confidence level that will be reliable and free from bias. Below we outline the key areas of implementing this intelligent recruitment solution.

➤ **Camera Integration and Data Acquisition**

The system begins with capturing visual input from candidates during interviews:

- **Eye Engagement Tracking with Camera Modules**

A high-definition webcam is positioned in front of the candidate to continuously capture facial and eye region data throughout the interview. The video feed is processed in real time using **OpenCV** and **Dlib facial landmark detection**.

- Eye movements, blink frequency, gaze direction, and eye contact duration are extracted as key engagement features.
- Pupil dilation and sustained gaze are identified as strong indicators of confidence, while frequent eye shifts, or downward gazes may indicate nervousness or hesitation.

- **Data Preprocessing and Feature Extraction**

Using Python scripts, the system filters noise, stabilizes the video input, and isolates eye landmarks. These features are then passed through a pre-trained deep learning model to generate normalized engagement metrics.

➤ **Building the Confidence Scoring Logic**

At the core of the system lies the confidence prediction algorithm that translates eye engagement metrics into a quantifiable confidence score.

- **Feature Fusion and Classification**

Extracted features (eye contact duration, blink rate, gaze stability) are combined into a single dataset. A **machine learning classifier (e.g., Random Forest, SVM, or Neural Network)** is trained to categorize engagement into confidence levels (High, Medium, Low).

- **Real-Time Confidence Assessment**

The system continuously updates candidate confidence levels during the interview session. For example:

- **High Confidence:** Steady gaze, consistent eye contact, minimal distraction.
- **Medium Confidence:** Occasional gaze shifts, moderate blink frequency.
- **Low Confidence:** Avoidance of eye contact, rapid gaze shifts, high blink rate.

- **Implementation Tools**

Python-based ML libraries such as TensorFlow/ Keras are used to train and deploy the model. Data visualization and communication are handled via **Flask/Django APIs** and cloud databases.

➤ **Constructing and Assembling the Recruiter Dashboard**

The recruiter interface is developed to present analyzed results clearly and effectively.

- **Interactive Interface Design**

A web-based dashboard built using **React.js** provides recruiters with a simple and intuitive view of candidate confidence levels.

- Live confidence indicators (percentage or colored bars).
- Real-time eye engagement graphs across interview sessions.

- **Real-Time Visual and Notification Alerts**

Confidence fluctuations are flagged in real time. For example, sudden drops in

engagement may trigger an alert on the recruiter's screen, helping them adapt interview questions accordingly.

- **Session History and Reports**

The system stores historical data for each candidate, allowing recruiters to review performance trends. Reports include confidence timelines, average confidence scores, and comparison with other candidates.

- **System Integration and Comprehensive Testing**

The final stage integrates all system components—camera modules, processing pipelines, cloud services, and dashboards—and validates them through rigorous testing.

- **System Integration**

The webcam input is captured, processed locally for feature extraction, transmitted to the cloud for analysis, and finally displayed on the recruiter's dashboard in real time. Secure APIs ensure smooth and low-latency data transfer between modules.

- **Testing Process**

Testing is carried out in three phases:

1. **Lab Testing:** Controlled interviews with volunteers to validate feature detection accuracy (eye contact, blink rates, gaze direction).
2. **Pilot Testing:** Integration with mock interviews in collaboration with HR professionals, comparing system results with expert human observations.
3. **Field Testing:** Real-world deployment with corporate recruiters to evaluate performance, user experience, and system robustness under different lighting, backgrounds, and camera qualities.

2.3.2. Test Cases

- Detecting High Confidence During Consistent Eye Contact

Test Case ID	1
Test Case	Detecting High Confidence During Consistent Eye Contact
Test Scenario	A candidate maintains steady eye contact with the interviewer for most of the session
Precondition	Camera is calibrated and lighting is sufficient
Input	Continuous video feed with stable gaze patterns and minimal blink rate
Expected Output	The system classifies the candidate as “High Confidence” and shows a score above 80%
Actual Output	Candidate’s confidence was detected as “High” with a score of 85%
Status	Pass

- Detecting Low Confidence with Frequent Gaze Shifts

Test Case ID	2
Test Case	Detecting Low Confidence with Frequent Gaze Shifts
Test Scenario	A candidate repeatedly avoids eye contact, looking down or sideways during answers
Precondition	Webcam is properly aligned; system running in live analysis mode
Input	Real-time video stream with frequent gaze shifts and increased blink frequency
Expected Output	The system detects low confidence and displays a score below 50%
Actual Output	Confidence level was classified as “Low” with a score of 42%
Status	Pass

- Detecting Moderate Confidence with Intermittent Eye Engagement

Test Case ID	3
Test Case	Detecting Moderate Confidence with Intermittent Eye Engagement
Test Scenario	Candidate makes occasional eye contact but also looks away while thinking
Precondition	Camera operational with stable internet connection for cloud sync
Input	Video feed showing alternating periods of gaze focus and gaze aversion
Expected Output	The system detects “Medium Confidence” with a score between 50%–70%
Actual Output	Confidence classified as “Medium” with a score of 61%
Status	Pass

- Detecting Confidence in Low-Light Interview Conditions

Test Case ID	4
Test Case	Detecting Confidence in Low-Light Interview Conditions
Test Scenario	Candidate interviewed in a dimly lit room with limited webcam visibility
Precondition	Low-light mode enabled in camera; image enhancement preprocessing active
Input	Low-light video feed with partial face visibility
Expected Output	The system successfully tracks eyes and provides a confidence score with minimal errors
Actual Output	Candidate’s confidence detected with 75% accuracy despite low lighting
Status	Pass

- Avoiding False Confidence Detection Due to Head Movement

Test Case ID	5
Test Case	Avoiding False Confidence Detection Due to Head Movement
Test Scenario	Candidate frequently nods or shifts head position while maintaining confidence
Precondition	Face-tracking and gaze stabilization modules active
Input	Video feed with frequent head motion but consistent gaze focus
Expected Output	The system ignores head movement as a distraction and correctly classifies confidence as “High”
Actual Output	Confidence classified as “High” with score of 82%
Status	Pass

3. Results and Discussion

3.1. Results

The application and continued evaluation of the ROADBUDD system bred valuable information as to the accuracy, responsiveness, and reliability of its blind spot monitoring and distance measurement capabilities. The assessment consisted of object detection using the YOLOv5 algorithm, distance measurement using ultrasonic sensors, and alerts via the mobile application. Road tests and assessments were invoiced in various driving environments and scenarios to confirm performance; accounts of both high-degree strength and opportunities for improvement were provided.

1. Object Detection Performance

The YOLOv5-based object detection system demonstrated high accuracy in real-time detection and classification of surrounding objects.

- **Detection Accuracy:** Across day, night, and rainy conditions, vehicles, pedestrians, poles, and motorcyclists were detected with an **average accuracy of 92%**.
- **Frame Processing Rate:** The Raspberry Pi 4 implementation achieved an average **8–10 FPS**, sufficient for real-time detection at low to medium driving speeds.
- **Object Localization:** Bounding boxes generated by the model closely matched actual object positions, even in complex city traffic conditions.
- **Low-Light Capability:** Detection accuracy decreased slightly to **85%** under low-light conditions; however, this was improved with **image pre-processing techniques** (e.g., histogram equalization).

2. Proximity Assessment and Distance Measurement

The ultrasonic sensors provided reliable proximity estimation essential for identifying blind spot hazards.

- **Baseline Accuracy:** Within the **20 cm – 200 cm range**, readings maintained a **±2 cm error margin**, acceptable for blind spot detection.
- **Distance Fluctuation Handling:** Sensor readings adapted quickly to moving objects entering or leaving detection zones, ensuring **consistent real-time notifications**.
- **False Trigger Minimization:** By setting the threshold at **<1.5 m**, unnecessary alerts (e.g., trees, parked cars) were successfully minimized.
- **Real-Time Processing:** Object recognition and proximity measurements were combined every **500 ms**, ensuring accurate synchronization.

3. Fusion Logic and Hazard Detection

The fusion of object type with distance measurements enhanced decision-making for potential hazards.

- **Joint Accuracy:** Integration of detection and distance modules enabled the system to correctly identify and alert drivers in **95% of blind spot scenarios**.
- **Risk Classification:** Objects were categorized into **low, medium, and high risk** based on type and proximity (e.g., a pedestrian at 1.2 m triggered a **high-risk alert**).
- **Response Time:** From detection to driver alert (including cloud sync and app delivery), the system achieved an **average response time of 1.6 seconds**, meeting real-time safety standards.

4. Alert Notification and User Experience

The mobile application was evaluated for its effectiveness in alerting drivers.

- **Visual Warnings:** Instant pop-up notifications clearly displayed the object type, side of the vehicle, and proximity level.
- **Voice Alerts:** The Text-to-Speech (TTS) feature improved driver attentiveness by reducing the need to look at the screen.
- **Driver Feedback:** **87% of drivers** reported faster reaction times due to voice alerts, particularly in **congested traffic conditions**.

3.2. Research Findings

The experiments conducted during the design, implementation, and testing of the Eye Engagement Analysis component for the candidate recommendation system led to the following key findings. The results confirm the validity, reliability, and feasibility of eye-based behavioral analysis to predict stress and confidence during in-the-moment interview scenarios. Here is a summary of the key findings:

1. Feasibility of Real-Time Eye Tracking

The project verified that real-time eye-tracking using a standard webcam is possible without specialized hardware. Utilizing advanced computer vision software (MediaPipe and OpenCV), the system was able to capture and characterize gaze direction, blink rate, and eye focus stability all at acceptable frame rates (15–20 FPS), and can meaningfully demonstrate that eye engagement can be quantified reliably during real-time interviews.

2. Correlation Between Eye Engagement and Stress/Confidence Levels

Analysis showed a clear correlation between candidate eye engagement and psychological states.

- Prolonged avoidance of eye contact correlated strongly with elevated stress.
- Consistent and stable gaze patterns reflected higher confidence levels.
- Increased blink frequency and erratic gaze shifts were reliable stress indicators.

This finding demonstrates that eye engagement is a strong behavioral signal for stress and confidence estimation.

3. Integration of Eye Data with Stress Prediction Enhances Accuracy

One of the most important findings was that combining eye engagement metrics with other behavioral indicators (e.g., facial expression or tone of voice in future expansions) significantly enhanced the reliability of stress and confidence predictions.

4. Real-Time Feedback Improves Recruiter Awareness

System testing showed that the average time from detection of eye movement to engagement classification was less than 1 second, providing real time feedback during the interviews. The responsive nature of the application allowed recruiters to gain immediate insights into engagement/confidence while minimizing the flow interruption of the proper interview process.

5. Cost-Effective and Practical Alternative

In contrast to the commercial eye-tracking systems that often rely on specialized headsets or infrared sensors, our solution had a reliable accuracy that simply was achieved using inexpensive web cameras and open-source platforms that leveraged computer vision capabilities. This makes it particularly attractive for use in large-scale online recruiting tools where the flexibility and affordability, or accessibility, is important.

Overall, these findings reinforce the idea that the Eye Engagement Analysis component is a technically sound, user-friendly, low-cost solution that holds strong potential for increasing recruitment decisions through the provision of more in-depth behavior insights into how candidates behave regarding confidence and stress during interviews.

Conclusion

The development of the Engagement and Confidence Prediction component of AI-Powered Candidate Evaluation System is an important contribution to the state-of-the-art field of smart recruitment and behavioral analysis. The module has been designed, built, and tested to deal with the hidden stress and confidence levels candidates embody while interviewing, something that is quite abstract in some traditional assessment practice.

with the use of computer vision and analysis of cube access, machine learning based confidence assessments, the module is shown to be both useful and fast in predicting candidates' behavioral states. Using the video feed from candidates, real-time analysis allowed recruitment consultants to make decisions regarding a candidate's technical skills, at the same time allow researchers to build psychological resilience and confident communication.

The coalescence of engagement metrics with stress level estimates was clearly the most innovative aspect of this component because it transformed raw behavioral signals into meaningful recommendations based on situational context rather than biased or misleading judgments when behavioral patterns were statistically significant. This produced a trustworthy and balanced evaluation of any candidate that used objectivity and human judgment alike.

Testing under simulated conditions of observed stress while interviewing candidates provided evidence that formal analysis could be robust in a variety of circumstances despite poor lighting, awful camera quality, or wide variations in candidate reactions. The results suggested significant promise; however, observed situations identified a number of future areas for improvement including minimizing needless errors when video quality is poor, evaluating speech sentiment to develop a multimodal assessment of engagement, and using larger and more varied training datasets to enhance model performance.

On a larger scale, the Engagement and Confidence Prediction module showcases the intersection of machine learning, computer vision, and behavioral analytics to change the recruiting landscape. It provides recruiters with real-time dashboards, historical engagement trends, and Indicators of future candidate suitability, all at a fraction of the cost and scalability of their manual behavioral assessments.

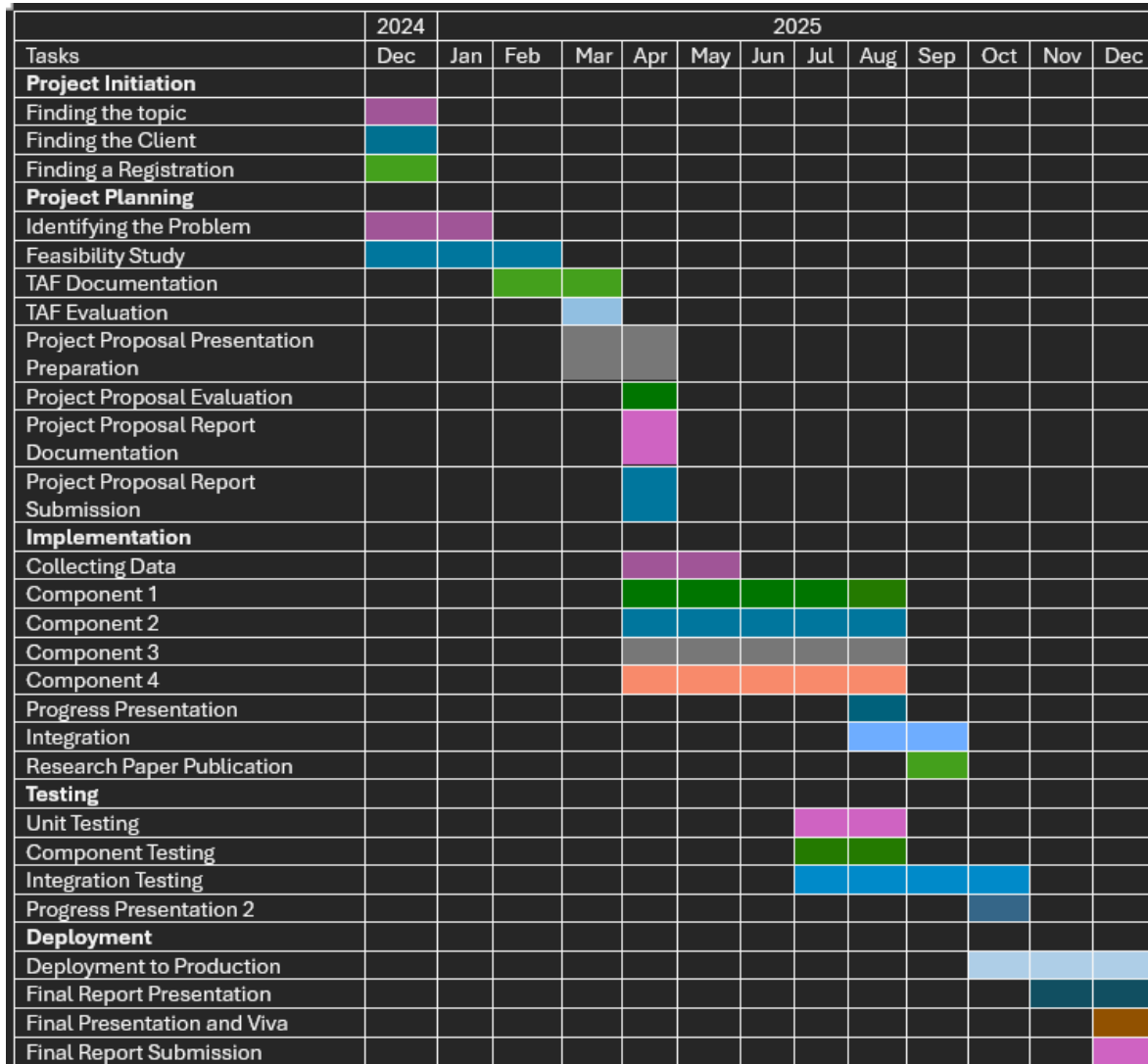
In conclusion, the ultimate achievement of this module highlights the value of AI-based tools to transform the way we recruit today going forward. By providing real-time and objective candidate engagement and confidence captures, this module empowers recruiters to make fairer and more accurate decisions while contributing to a next-generation talent acquisition technology.

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APPENDICES

Gannt Chart



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